

Learning Objective 1.5: I can identify and distinguish the reactive near-field, radiating near-field, and far-field regions and calculate the boundaries for a given antenna.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Three terms in one equation.** The exact E_θ of an infinitesimal dipole is a sum of three pieces whose relative sizes are set entirely by kr :

$$E_\theta \propto \underbrace{\frac{1}{r}}_{\text{radiation}} + \underbrace{\frac{1}{kr^2}}_{\text{induction}} + \underbrace{\frac{1}{k^2r^3}}_{\text{electrostatic}}$$

Throughout this question the antenna operates at $f = 915\text{MHz}$.

- (a) The innermost region is called *reactive*. Justify that word in one or two sentences, using what the complex Poynting vector does there.

- (b) Normalizing the radiation term to 1, give the relative sizes of the three terms at $kr = 0.1$ and at $kr = 10$, and name the dominant term in each case.

$kr = 0.1$:

$kr = 10$:

- (c) Find the distance at which the radiation and induction terms are equal. Give it in wavelengths and in centimetres at 915MHz.

$r =$

- (d) A colleague measures this antenna's gain with a probe held 2 cm away. In two sentences, say what is wrong with the number he reports.

2. **An electrically large antenna.** An X-band ground-station reflector has diameter $D = 3.0\text{m}$ and operates at $f = 8\text{GHz}$. The boundaries are $r_{\text{react}} = 0.62\sqrt{D^3/\lambda}$ and $r_{\text{ff}} = 2D^2/\lambda$.

(a) Find λ and the electrical size D/λ . Is this antenna electrically large?

 $\lambda =$ $D/\lambda =$

(b) Compute both region boundaries.

 $r_{\text{react}} =$ $r_{\text{ff}} =$

(c) The only outdoor range on base is 150m long. Which region does that put the source in, and what fraction of the far-field distance is it?

region =

fraction =

(d) You need this dish's sidelobe pattern and you cannot build a 480m range. Give two ways out and state what each costs.

3. **An electrically small antenna.** A 915MHz ISM whip on a sensor node has largest dimension $D = 8$ cm. Recall $\lambda = 0.328$ m at this frequency.

(a) Compute D/λ , $2D^2/\lambda$, and $0.62\sqrt{D^3/\lambda}$.

$$D/\lambda =$$

$$2D^2/\lambda =$$

$$0.62\sqrt{D^3/\lambda} =$$

- (b) Your $2D^2/\lambda$ came out *smaller than a wavelength*. Explain in two or three sentences why that number is not the far-field boundary here, and say what replaces it.

- (c) Compute $\lambda/2\pi$ for this antenna, and state the distance at which you would actually stand to measure its pattern (use the $r \geq 5\lambda$ practical rule).

$$\lambda/2\pi =$$

$$r_{\text{meas}} =$$

- (d) You are offered a 3m anechoic chamber to characterize this whip. Is it long enough? Name the error source that actually limits the measurement, given your answer to (c).

4. **Why ranges are long: the $\pi/8$ criterion.** A source a distance r away is farther from the *edge* of an aperture of size D than from its centre, by a path difference $\Delta \approx D^2/(8r)$. The agreed tolerance is $\Delta \leq \lambda/16$. Take the lesson's reflector: $D = 1.2\text{m}$ at $f = 10\text{GHz}$, so $\lambda = 0.03\text{m}$.

- (a) At $r = 30\text{m}$, find the path difference Δ and the phase error it produces across the aperture, in degrees.

$\Delta =$

$\Delta\phi =$

- (b) Impose $\Delta \leq \lambda/16$ and solve for the minimum range. Show that the $\pi/8$ tolerance is what produces the familiar $2D^2/\lambda$.

$r_{\text{ff}} =$

- (c) Your chamber is the 30m one from part (a) and the program will not fund a 96m outdoor range. State what the 72° of residual phase error does to the measured pattern, and decide whether you would accept the 30m data for a *gain* check, for a *beamwidth* check, and for a *sidelobe* check.

- (d) Why is 22.5° "good enough"? Answer in one or two sentences, saying what would change if the tolerance were tightened to $\pi/16$.

5. **Two antennas on one airframe.** A 3GHz ($\lambda = 0.1\text{m}$) installation carries a half-wave dipole ($D = \lambda/2 = 0.05\text{m}$) and a 2.0m reflector, mounted 10m apart.

(a) Inside $2D^2/\lambda$ the field is genuinely radiating – energy is leaving and never coming back – and yet the *pattern* still changes as you walk outward. Explain why, in words.

(b) Compute $2D^2/\lambda$ for each antenna.

dipole =

dish =

(c) At the 10m separation, which region is each antenna in with respect to the other?

dish:

dipole:

(d) The EMC engineer wants to predict the coupling between the two with Friis. Is that legitimate here? Say what you would do instead, and what it costs.

Documentation: _____